MILITARY SPECIFICATION

CARTRIDGE, 152MM: CANISTER, XM625E2 LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers the loading, assembling and packing for one type of cartridge designated as Cartridge, 152MM, Canister, XM625E2.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-223	-	Powder, Black
MIL-A-2550	-	Ammunition and Special Weapons, General Specification for
MIL-I-45607	-	Inspection Equipment, Supply and Maintenance of
MIL-I-50867	-	Ignition Element, Electric, Loading Assembling and Packaging

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105).
MIL-STD-109	-	Quality Assurance Terms and Definitions
MIL-STD-286	-	Propellants, Solid: Sampling, Examination and Testing
MIL-STD-414	-	Sampling Procedures and Tables for Inspection by Variables for Percent Defective
MIL-STD-1168	-	Lot Numbering of Ammunition
MIL-STD-1235	-	Single and Multilevel Continuous Sampling Procedures and Tables for Sampling by Attributes

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DRAWINGS

ARMY

9212218	-	Igniter Assembly
9224909	-	Box, Packing, Ammunition, for Cartridge, 152MM, Canister, XM625E2
9221407	-	Container, Ammunition, Fiber, M580 for Cartridge, 152MM, Canister, XM625E2
9252590	-	Cartridge, 152MM, Canister, XM625E2
9260121	-	Cartridge Assembly for 152MM with Bag, Barrier, Elastomeric
9249028	-	Ignition Element, Electric

PUBLICATIONS

ARMY

ET-9252590 - Equipment Tabulation

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 General.-Material and parts shall be in accordance with applicable drawings and specifications.

3.2 Cartridge.-The cartridge shall comply with all requirements specified on drawing (dwg.) 9252590 and with all requirements specified in applicable specifications.

3.3 Moisture content of black powder in igniter assemblies.-The moisture content of black powder loaded into igniter bags (at the time of loading) shall not exceed 0.70 percent, when determined in accordance with MIL-P-223.

3.4 Pre-use inspection of cartridge case components.-The gap between the cartridge case body outside diameter and the cartridge case base inside diameter shall not exceed 0.010 inch for a depth of 0.0625 inch maximum (max) when inspected in accordance with 4.3.1.

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3.5 Ignition element circuit continuity.-The resistance of the ignition element after insertion into the cartridge base shall not be greater than 1.0 ohm when tested in accordance with 4.3.2.

3.6 Assembly torque.-The threaded joint shall be tightened to the assembly torque specified on the applicable drawing while thread sealing compound is wet (see 4.3.3 and 6.5).

3.7 Moisture content of cartridge case.-At the time of assembly of elastomeric barrier bag to the cartridge, the average moisture content of the case body and case base shall not exceed 1.5 percent and no individual reading shall exceed 2.0 percent when tested in accordance with 4.2.3.5.

3.8 Drying time.-After assembly of cartridge case base to cartridge case body, the cartridge shall be cured in the conditioning bay in accordance with the drying time specified on the drawing.

3.9 Workmanship

3.9.1 All parts shall be free of cracks and burrs; and chips, dirt, grease, rust, or other foreign material. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agents.

3.9.2 The painting and marking shall be restored by the assembling contractor on any components furnished to him painted and marked, when such paintings have been marred or obliterated. (Bare spots in painting, other than marking, to total not more than 1/4 square inch in aggregate, are allowable.)

3.10 Proving ground.-The cartridge shall comply with the following requirements when tested as specified in 4.3.6.

3.10.1 Functioning.-The cartridge shall function satisfactorily after leaving the gun and prior to reaching a distance of 7.6 meters from the gun muzzle.

3.10.2 Security.-There shall be no evidence of metal parts separation prior to leaving the gun tube as evidenced by damage to the gun tube.

3.10.3 Residue.-The cartridge case components shall function satisfactorily without excessive or smoldering residue (see 4.3.6.2).

3.10.4 Pressure.-No individual pressure obtained when fired at 70°F shall be greater than 39,300 pounds per square inch (psi) or less than 33,000 psi.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define terms used herein. The provisions of Specification MIL-A-2550 shall apply.

4.1.1 Submission of product.-At the time the completed lot of product is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified within this specification.

b. Drawing, specification number and date, together with an identification and date of changes.

c. A statement that all material purchased by the contractor meets requirements when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.

d. Number of items in the lot.

e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 Inspection provisions

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4.2.1 Lot formation.-The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, in accordance with the same specification revision and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to the process, specification, or drawing, not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each inspection lot of loaded complete round cartridges shall contain::

a. Ignition element assemblies of one lot interfix number from one manufacturer.

b. Cartridge case bodies and bases of one lot interfix number from one manufacturer.

c. Canister metal parts (excluding case locking rings) of one lot interfix number from one manufacturer.

d. Propellant charges of one propellant lot from one manufacturer.

e. Black powder of one lot from one manufacturer.

f. Case locking rings produced to the same drawing, drawing revision, specification, and specification revision by one manufacturer.

g. Cartridge case mounting rings produced to the same drawing, drawing revision, specification and specification revision by one manufacturer.

4.2.2 Examination.-The examinations listed in the following subparagraphs shall be performed on inspection lots (see 4.2.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups

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of major and minor defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. When this method is elected, it shall be applied to all of the major and minor defects listed in the applicable sub-paragraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.2.4.

4.2.2.1 Assembly of Ignition Element (see dwg. 9252590)

Categories	Defects	Method of Inspection	Code No. (see 6.2)
Critical:			
1.	Ignition element inserted inversely	Test	01001
Major:	AQL 0.65 percent		
101.	Ignition element loose	Manual	01002
102.	Ignition element below flush tolerance	Gage	01003
103.	Ignition element above flush	Straight-edge	01004
104.	Adhesive missing from outside diameter of ignition element or inside diameter of hole in base (prior to assembly)	Visual	01005
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.9)	Visual	01006

4.2.2.2 Igniter assembly, prior to loading of black powder (see dwg. 9212218).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined.			
Major:	AQL 0.40 percent		
101.	Incorrect weight of igniter charge	Gage	02001
102.	Igniter bag torn, ripped or cut	Visual	02002
Minor: None defined.			

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4.2.2.3 Assembly of igniter components (see dwg. 9252590)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Tube cracked or broken	Visual	03001
102.	Igniter assembly missing or inverted	Visual	03002
103.	Closing plug missing	Visual	03003
Minor:	AQL 1.0 percent		
201.	Adhesive applied incorrectly to tube	Visual	03004
202.	Evidence of poor workmanship (see 3.9)	Visual	03005

4.2.2.4 Assembly, after assembling cartridge case (see dwg. 9252590).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	100 percent inspection		
101.	Thread sealant missing from case locking ring threads	Test	04001
102.	Case mounting ring not firmly seated	Manual	04002
103.	Incorrect number of case mounting rings assembled	Visual-Manual	04003
Minor:	None defined.		

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4.2.2.5 Assembly, prior to assembling cartridge case base and propellant charge (see dwg. 9252590).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	100 percent inspection		
101.	Case locking ring fails torque requirement (see 3.6)	Visual	05001
102.	Cracks in base area of cartridge case (if flared)	Visual	05002
103.	Propellant charge damaged (prior to assembly)	Visual	05003
104.	Case cracked or broken near mounting ring	Visual	05004
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.9)	Visual	05005

4.2.2.6 Cartridge (see dwg. 9252590).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	100 percent inspection		
101.	Incomplete, insufficient or incorrect application of adhesive to base (prior to assembling)	Visual	06001
102.	Propelling charge missing (prior to assembling base)	Visual	06002
103.	Excess adhesive not removed after assembly of base	Visual	06003
104.	Distance from datum diameter on rotating band to base of case	Gage	06004
105.	True position of ignition element with canister	Gage	06005
106.	Improper or incomplete coating of lubrication on rotating band or lubricant not dry to touch	Manual-Visual	06006
107.	Closing torque on clamp (see 6.7)	Visual	06007
108.	Rotating band damaged	Visual	06008
109.	Cartridge case-canister gap incorrect	Gage	06009
110.	Cartridge case sidewall-base gap incorrect	Gage	06010

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Major:	AQL 0.40 percent		
111.	Assembly time (see dwg. 9252590).....	Test	06011
Minor:	AQL 1.00 percent		
201.	Marking incorrect or unidentifiable including color coding.....	Visual	06012
202.	Evidence of poor workmanship (see 3.9).....	Visual	06013

4.2.2.7 Unsealed fiber container, prior to loading of cartridge (see dwg. 9221407).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	100 Percent Inspection		
101.	Barrier bag (elastomeric) torn, ripped or missing from cartridge.....	Visual	07001
102.	Neck tube or ring missing.....	Visual	07002
103.	Desiccant bag, broken, missing or not stitched into ring.....	Visual	07003
104.	Sleeve (two halves) missing or improperly assembled.....	Visual	07004
Major:	AQL 0.65 percent		
106.	Round cannot be removed by hand (after inserting).....	Manual	07006
107.	Cushion missing.....	Visual	07007
108.	Discs between cushion and cartridge missing.....	Visual	07008
109.	Support or related components including pad missing.....	Visual	07009
Minor:	None defined.		

4.2.2.8 Sealed fiber container (see dwg. 9221407)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Container damaged or cut through all impregnated layers.....	Visual	08001
102.	Sealing strip applied improperly...	Visual	08002
103.	Metal end loose or distorted.....	Visual-Manual	08003

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Categories	Defects	Method of Inspection	Code No.
Minor:	AQL 2.5 percent		
201.	Contents loose	Manual	08004
202.	Tear tab length inadequate	Visual	08005
203.	Gap between body and cover in excess of 1/8 inch	Visual	08006
204.	Marking misleading or unidentifiable	Visual	08007
205.	Evidence of poor workmanship (see 3.9)	Visual	08008

4.2.2.9 Sealed wooden packing box (see dwg. 9224909).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	100 percent inspection		
101.	Barrier bag improperly sealed, torn, or missing prior to placing round in box	Visual	09001
102.	Contents loose (prior to sealing)	Manual	09002
Major:	AQL 0.65 percent		
103.	Hardware or strapping missing, broken or loose, or improperly located	Visual-Manual	09003
104.	Box damaged exposing contents	Visual	09004
105.	DOD code marking missing, incorrect or unidentifiable	Visual	09005
Minor:	AQL 0.65 percent		
201.	Car seal missing, unsealed or improperly positioned	Visual	09006

4.2.3 Testing

4.2.3.1 Check test for deterioration of ignition elements (dwg. 9249028) (see 3.5), Major defect, Code No. 10001.-If the total elapsed time between original acceptance of an assembled ignition element lot and assembly of that lot into the ammunition components with which it is to be issued exceeds two (2) years, or if the ignition elements have been subjected to adverse conditions at any time since previous tests, or if more than six (6) months have elapsed since a previous check test, the ignition element lots shall be subjected to, and must satisfactorily pass, the check test for deterioration specified in MIL-I-50867, prior to use (see 6.4).

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4.2.3.2 Moisture content of black powder (see 3.3), Major defect, Code No. 10002.-The contractor shall provide adequate controls to assure the compliance of each lot with the applicable requirement. For verification, and just prior to the start of each eight hours loading production, the contractor shall test one sample of black powder selected from that quantity, or sub-lot of powder which is to be used in the eight-hour loading production of ignition element base subassembly. Composite samples shall not be used. Testing shall be in accordance with MIL-P-223 and if the moisture content of the sample exceeds the requirement, and loading has not begun, the sub-lot of powder represented by the sample shall be rejected. If powder from a sub-lot with excessive moisture has been used in loading, the remaining unloaded powder and all assemblies containing the non-conforming powder shall be rejected.

4.2.3.3 Pre-use inspection of cartridge case components.- Major defect, Code No. 10003.-For this inspection a "lot" shall be defined as that quantity of cartridge case bodies and bases which are scheduled for use in eight hours of loading. Prior to use of each "lot", fifteen bodies and bases from the "lot" shall be inspected in accordance with 4.3.1. Failure of any sample shall require flaring of the failing sample(s) and reinspection. If reinspection indicates compliance with the applicable requirement (see 3.4), the remainder of the "lot" shall be flared prior to use. If reinspection still indicates rejection, the "lot" shall not be used in loading.

4.2.3.4 Ignition element circuit continuity (see 3.5), Major defect, Code No. 10004 (100 percent inspection).-Each ignition element base subassembly shall be tested for continuity in accordance with 4.3.2 using equipment in accordance with 4.2.4. Any subassembly failing to comply with the requirements shall be rejected and removed from the lot.

4.2.3.5 Moisture content of cartridge case (see 3.6) - Major defect - Code No. 10005.-The first cartridge loaded during each four hours of production shall be a test cartridge case-canister assembly (use no adhesive, no ignition element and no igniter assembly) and propelling charge. The test cartridge (representing the subsequent four hours of production) shall be included on the first skid of production cartridges placed in the conditioning bay. One hour prior to completion of final drying of the first skid of production cartridges, the test cartridge case-canister assembly shall be subjected to a moisture test in accordance with MIL-STD-286B, Method 101.4. The samples required for the test shall be prepared as follows: Two 0.025 gram (gm) (approximate) samples shall be sectioned from the case body at random locations and two 0.025 gm (approximate) samples shall be sectioned from the case base. The temperature control shall be set at 100 degrees Centigrade with a timer setting of 15 minutes. Cartridges represented by the samples shall remain identified until successful performance during the moisture content

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test is indicated. If the samples fail to comply with the applicable requirement, the four hours of production cartridges represented by the failing samples shall remain in the conditioning bay for an additional 4 hours ± 1/2 hour. One hour prior to completion of the additional curing time a production cartridge shall be randomly selected and subjected to testing as specified above. Failure of these samples to comply with the applicable requirement shall reject the four hours of production cartridges which the samples represent.

4.2.3.6 Drying time - Major defect, Code No. 10006.-100 percent inspection.-Each skid of assembled cartridges shall be visually inspected in accordance with 4.3.4 for drying time in the conditioning bay. If any skid document indicates time in the conditioning bay lower than the minimum total requirement, the skid of cartridges represented by the document shall be placed back in the conditioning bay until compliance with the applicable requirement (see 3.8) is indicated. If any skid document indicates time in the conditioning bay exceeding the maximum total requirement (see 3.8), the skid of cartridges represented by the document shall be rejected (see 6.9).

4.2.3.7 Pull test of cartridge case base (see dwg. 9252590).

4.2.3.7.1 Production - Major defect, Code No. 10007.-This test shall be conducted on a 100 percent basis. Any cartridge which fails to comply with the applicable requirement shall be removed from the lot. Testing shall be as specified in 4.3.5.

4.2.3.7.2 Destruction (for information only).-Inert pull test cartridges (three total) shall be loaded at random intervals with production cartridges in a lot. The three inert pull test cartridges shall be assembled with only the cartridge case body, mounting ring, locking ring (no thread sealant) and cartridge case base using adhesive and shall be conspicuously identified. After final drying, the three pull test cartridges shall be subjected to a pull test to destruction of the cartridge case base. Results as to force required to cause separation shall be forwarded to Picatinny Arsenal, SMUPA-ND2.

4.2.3.8 Profile and alignment (see dwg. 9252590) - Major defect, Code No. 10008.-The cartridge shall be gaged 100 percent for profile and alignment and the maximum allowable force for extracting the round from the chamber shall be applied. Any cartridge which fails to comply with the applicable requirement shall be classed defective and removed from the lot. The test shall be performed using equipment in accordance with 4.2.4.

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4.2.3.9 Proving ground.-The following tests shall be performed as specified in 4.3.6. (See 6.12 for Test Summary)

4.2.3.9.1 Functioning (see 3.10.1) - Major defect - Code No. 10009.

4.2.3.9.1.1 Beginning with the first lot produced and continuing until three consecutive lots have complied with the applicable requirements specified, 50 cartridges shall be selected from each lot for this test. If two or more cartridges fail to meet the applicable requirements, the lot shall be rejected.

4.2.3.9.1.2 After three consecutive lots have complied with the criteria of 4.2.3.9.1.1, 30 cartridges shall be selected from each lot for this test. If two or more cartridges fail to meet the applicable requirements, the lot shall be rejected.

4.2.3.9.2 Security (see 3.10.2) - Critical defect - Code No. 10010.-All rounds fired in this ballistic test shall be observed for conformance to security requirements. If any round fails to comply with the applicable requirements, the lot shall be rejected.

4.2.3.9.3 Residue

4.2.3.9.3.1 Beginning with the first lot produced and continuing until three consecutive lots have complied with the applicable requirements, fifty (50) sample cartridges fired during the proving ground acceptance tests, shall be observed for excessive residue (Major defect, Code No. 10011) and smoldering residue (Special defect, see 6.13, Code No. 10012). The lot shall be rejected for any smoldering residue or evidence of two or more rounds exhibiting excessive residue (see 6.11).

4.2.3.9.3.2 After three consecutive lots have met the criteria of 4.2.3.9.3.1, thirty (30) sample cartridges fired during the proving ground acceptance tests, shall be observed for excessive and smoldering residue. The lot shall be rejected for any smoldering residue or evidence of two or more rounds exhibiting excessive residue (see 6.11). Upon rejection of any one lot for presence of residue, tightened inspection in accordance with 4.2.3.9.3.1 as well as 4.2.3.9.1.1 shall be instituted until three consecutive lots have again complied with the applicable requirements.

4.2.3.9.4 Pressure - Major defect, Code No. 10013.-Five randomly selected sample cartridges shall be selected for this test. If any sample fails to meet the requirements of 3.10.4, the lot shall be rejected.

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4.2.4 Inspection Equipment.-Equipment Tabulation Number ET-9252590 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in 6.3.

4.2.4.1 Government rights to documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements of the ET may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.2.4.2 Supply and maintenance.-Supply and maintenance of the equipment listed in the ET shall be in accordance with Specification MIL-I-45607.

4.2.4.3 Government use of contractor's inspection equipment.-The contractor shall make available to the Government all inspection equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment and verification of its accuracy, shall be supplied by the contractor for the performance of examination or test by the Government.

4.3 Test methods and procedures

4.3.1 Pre-use inspection of cartridge case components.-The cartridge case base shall be assembled to the cartridge case body (without glue) and the base clamp assembled and torqued to the required value. A 0.010 inch thick by 0.375 inch wide feeler gage shall be used to check the gap between the outside diameter of the cartridge case body and the inside diameter of the cartridge case base. The gap shall be checked at four locations, 90 degrees apart. The gage shall not enter to a depth exceeding 0.0625 inch at any point.

4.3.2 Ignition element circuit continuity.-The ignition element after insertion into the cartridge case base shall be tested for resistance by the following method: A probe shall be inserted through the test holes of the base and ignition element to the inner sleeve of the ignition element and the circuit completed by connecting a lead from the probe to the outer cup of the ignition element. An ohmmeter shall be attached and the resistance read to the nearest 0.1 ohm. No more than 10 milliamperes current shall be induced in the circuit during this test.

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4.3.3 Assembly torque.-The torque shall be applied while holding the parts immediately adjacent to the joint so as to avoid possible loosening of any other joints.

4.3.4 Drying time.-For each skid of cartridges, the document representing the skid shall be annotated with the time and date the skid of cartridges entered the conditioning bay and the time and date the skid of cartridges was removed from the conditioning bay.

4.3.5 Pull test of cartridge case base

4.3.5.1 Production.-This test shall be performed after the bond at the cartridge case body/base joint has had sufficient time to set. The cartridge shall be held in a suitable test fixture clamping on the canister body. The test apparatus shall apply a longitudinal load of 150 pounds distributed over 360 degrees around the lip of the cartridge case base. Observation shall be for separation of the cartridge case base from the cartridge. The method of test should not damage the ammunition components and paint or markings damaged during the test shall be restored. Cartridges which meet the requirement may be returned to the lot provided they have not been damaged by the method of test.

4.3.5.2 Destruction.-This test shall be performed after the bond at the cartridge case body/base joint has had sufficient time to set. The cartridge shall be held in a suitable test fixture clamping on the projectile body. The test apparatus shall apply a longitudinal load distributed over 360 degrees around the lip of the cartridge case base. Observation shall be for the force required for separation of the cartridge case base from the cartridge.

4.3.6 Proving ground tests.-These tests shall be conducted at a Government proving ground in accordance with the applicable Acceptance Test Procedure. The sample cartridges shall be fired in a gun for which the cartridge is standard without a closed breech bore scavenger utilizing a 110 volt direct current (DC) power source (see 6.10) and observation shall be made for conformance to all the requirements.

4.3.6.1 Functioning and Security.-Adequate photographic coverage and visual observations shall be made to determine compliance with the applicable requirements. Adequate instrumentation shall be used to determine distance to function if requirements are not met.

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4.3.6.2 Residue.-Camera coverage of the breech chamber cavity, tube chamber and bore, shall be used to determine compliance with the smoldering residue requirement immediately after firing and upon opening of the automatically operated breech mechanism. Still photographs shall be taken of excessive residue and the residue shall be collected (see 6.11).

4.3.6.3 Pressure.-Chamber pressures shall be taken on each round fired and reported as maximum piezoelectric equivalent pressure.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging

5.1.1 Level A.-The cartridge shall be preserved and packaged in accordance with drawing 9221407. In addition, the container and desiccant bag (cover removed from body with one each desiccant bag placed inside the body) shall be dried in a circulating oven for a minimum of 8 hours at a temperature of 100 Degrees Fahrenheit (°F) + 10°F, 10 percent relative humidity max, prior to packing. If the container is not used within fifteen minutes after removal from the oven the container shall be closed tightly, with desiccant bag remaining inside. Final packing of the cartridge in the container shall be accomplished within one hour of the time the container was removed from the oven.

5.2 Packing

5.2.1 Level A.-The containers shall be packed in accordance with dwg. 9224909.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Data cards.-Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.

6.2 Inspection code numbers.-The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government. These code numbers are used to correlate the characteristics cited on Equipment Lists with the inspections listed in this specification. They should also be cited as reference on drawings of equipment designs submitted by the contractor (see 6.3.5).

6.3 Inspection equipment.-The contractor shall design inspection equipment as required by the inspection Equipment Lists (EL) referenced on the applicable ET in accordance with the instructions of paragraphs 6.3.1 through 6.3.5.

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6.3.1 Inspection equipment lists (EL).-Inspection equipment lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (Form SMUPA 1010) the numbers of drawings or Federal Stock numbers of existing equipment designs, or codes as indicated in paragraph 6.3.2. Design action required of the contractor is described in paragraphs 6.3.3 and 6.3.4. The contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.2 Inspection equipment list codes.-The inspection equipment as defined in 6.3.3 and 6.3.4 will be designated in the EL by the following codes:

CD - Contractor design on controlled contractor format and/or commercial equipment.
MU - Army design, mandatory for use.

6.3.3 Army designs.-Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications; mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an inspection equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.4 Contractor designs.-Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in

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accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to MIL-D-1000, Category E, Form 2. Legibility and reproducibility shall permit conventional making of clearly understandable, high contrast reproductions. Contractors shall submit two copies of final designs as a flat set. Designs shall be submitted for approval as specified in 6.3.5.

6.3.5 Submission of contractor designs.-All submitted designs shall contain a reference to the applicable five digit Code Number contained in Section 4 of this specification and the appropriate component or assembly drawing number and revision letter to which the specific design applies. Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND12. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract; submission shall be as directed by the contracting officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished normally within one month after receipt.

6.4 Cost of check test.-The contracting officer will arrange for the contractor to be reimbursed for the expense incurred in the performance of check tests. The test shall be conducted at Government expense without cost to the contractor who loaded the primers or to the contractor assembling the cartridge, and shall not constitute a basis for rejection against either contractor, except where deterioration has occurred as a direct result of carelessness in handling, storage, etc., permitted while the primer lot was under the jurisdiction of either contractor.

6.5 Torque application to case locking ring.-Due to the compressibility of the cartridge case material, during the assembly of the case body to the projectile, the case locking ring will lose its torque almost immediately. Inspection for compliance with the applicable torque requirement is therefore limited to witnessing the torque application.

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6.6 Gluer Qualification.-Since the gluing operation associated with the assembly of the cartridge case base to the cartridge case body is extremely important to overall cartridge case strength, it is recommended that personnel associated with this operation be carefully trained. A gluer qualification test is suggested. Properly assembled cartridge case body to base joints should be capable of withstanding in excess of 1000 pounds of force during pull testing.

6.7 Torque application to base clamp.-Due to the requirement prohibiting reapplication of assembly torque, inspection for compliance with the applicable torque requirement is limited to witnessing the torque application.

6.8 Moisture content of cartridge case base.-It is recommended that a verification of moisture content of cartridge case parts be conducted by the contractor upon their introduction to the production line to reduce the possibility of cartridge rejection due to excessive moisture content of the case at the time of the barrier bag assembly.

6.9 Documentation on drying time.-Documentation on drying time shall be retained until the lot which the documents represent has been finally accepted.

6.10 Misfires.-If a misfire not related to the weapon firing system is experienced during any phase of proving ground testing with a 110 volt DC power source, the misfired round(s) shall be held for investigation and the lot shall be referred for disposition.

6.11 Residue

6.11.1 Excessive residue.-Excessive residue shall be defined as residue large enough to prevent chambering or otherwise hinder functioning of a subsequent round. In addition, any single piece of residue larger than 1 inch by 2 inches should be considered excessive. Any collected residue and photographs shall be forwarded to Picatinny Arsenal, SMUPA-ND2, for evaluation.

6.11.2 Smoldering residue.-All movies of smoldering residue shall be forwarded to Picatinny Arsenal, SMUPA-ND2, for evaluation.

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6.12 Proving Ground Test Summary

Test	Sample Size		Requirements
Function & Residue	*50	30	a. Satisfactory function within specified distance (see 3.10.1) b. Security (see 3.10.2) c. Residue, excessive or smoldering (see 3.10.3) d. Distance to function when beyond specified distance for information (see 4.3.6.1) e. Misfires (see 6.10)
Pressure	5	5	a. Pressure at 70°F (see 3.10.4) b. Security (see 3.10.2) c. Misfires (see 6.10)
	*55	35	

* For first three lots or tightened inspection.

6.13 Special defects

6.13.1 Definition.-Special defect is a defect that judgement and experience indicate could, depending upon the degree of variance from the design requirement, result in hazardous or unsafe conditions for individuals using, maintaining or depending upon the product.

6.13.2 Disposition.-Upon investigation and analysis of lots exhibiting special defects by the responsible technical activity a recommendation of acceptance on waiver or rejection shall be made to the Director of Quality Assurance, U. S. Army Munitions Command. Final acceptance of such lots on waiver shall be made only by the Director of Quality Assurance, U. S. Army Munitions Command.

6.14 Submission of test data and ammunition data cards.-In addition to the normal distribution of records, when the cartridge is procured by the Army Munitions Command, one copy of all ballistic test data and ammunition data cards shall be forwarded to each of the following:Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND2, SMUPA-DC10, and SMUPA-KI, Dover, N.J.

Custodian:
ARMY-MU

Preparing Activity:
ARMY-MU

FSC: 1320-A

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 OCT 64 1426

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